In [190…

```python
"""
Homework 2 Who tends to exercise more?
Jackson School of Global Affairs

 Created by Ardina Hasanbasri for GLBL 5021

Additional reference code and data used:
Békés & Kézdi (2021) see more code below
https://gabors-data-analysis.com/

"""

# Note: Feel free to move this code to jupyter notebook if you prefer.

#----------------------------------#
# SETTING UP YOUR WORKSPACE #
# ----------------------------------#

# Feel free to add or delete
import os
import sys
import warnings
import numpy as np
import pandas as pd
import pyfixest as pf
import seaborn as sns
import matplotlib.pyplot as plt
import pyreadstat
from statsmodels.nonparametric.smoothers_lowess import lowess
from sklearn.metrics import mean_squared_error, r2_score, log_loss
import statsmodels.formula.api as smf
from typing import List
import marginaleffects as me
warnings.filterwarnings("ignore")

share = pd.read_csv("https://osf.io/download/kbjzp/")

#----------------------------------------------------------#
# Sample Selection and Creating New Data                   #
# ----------------------------------------------------------#

share = share.query("wave==4 & age>=50 & age<=64")

# Some reminder code to help explore data.
share["br015"].unique()
share["mar_stat"].value_counts()
share["bmi"].describe()
```

Out[190…

```
count    26254.000000
mean        25.195472
std          9.284344
min        -15.000000
25%         23.243801
50%         26.026175
75%         29.372396
max         88.376396
Name: bmi, dtype: float64
```

In [191…

```
share.index = share.mergeid
share = share.drop(columns=['mergeid'])
```

Part 1

1.1

In [192…

```
share.describe()
```

Out[192…

	wave	country	country_mod	int_year	int_month	female	age	eduyears_mod	sphus	br015	smoking	ever_smoked	income_pct_w4
count	26254.0	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000	26254.000000
mean	4.0	22.593205	349.048640	2011.005028	5.797364	0.562048	57.572903	10.149901	2.972538	2.252114	3.835796	2.528948	6.021025
std	0.0	8.230092	241.682133	0.236737	2.712384	0.496145	3.903833	6.681391	1.525931	1.879895	2.241819	3.172209	2.943781
min	4.0	11.000000	40.000000	2010.000000	1.000000	0.000000	50.000000	-15.000000	-15.000000	-15.000000	-15.000000	-15.000000	1.000000
25%	4.0	16.000000	203.000000	2011.000000	4.000000	0.000000	54.299999	8.000000	2.000000	1.000000	1.000000	1.000000	4.000000
50%	4.0	20.000000	250.000000	2011.000000	5.000000	1.000000	57.799999	11.000000	3.000000	2.000000	5.000000	1.000000	6.000000
75%	4.0	32.000000	616.000000	2011.000000	8.000000	1.000000	60.900002	14.000000	4.000000	4.000000	5.000000	5.000000	9.000000
max	4.0	35.000000	756.000000	2012.000000	12.000000	1.000000	64.000000	25.000000	5.000000	4.000000	5.000000	5.000000	10.000000

In [193…

```
cols_to_filter = share.columns.difference(['female'])
share_filtered = share[(share[cols_to_filter] > 0).all(axis=1)].dropna(subset=cols_to_filter)
```

In [194…

```
share_filtered.describe()
```

Out[194…

	wave	country	country_mod	int_year	int_month	female	age	eduyears_mod	sphus	br015	smoking	ever_smoked	income_pct_w4	2
count	23926.0	23926.000000	23926.000000	23926.000000	23926.000000	23926.000000	23926.000000	23926.000000	23926.000000	23926.00000	23926.000000	23926.000000	23926.000000	
mean	4.0	22.861364	342.346945	2011.001546	5.824459	0.558597	57.523869	11.432208	3.030594	2.34590	3.932208	2.872607	6.054585	
std	0.0	8.237887	237.699239	0.236567	2.713939	0.496565	3.902150	4.072558	1.071325	1.31931	1.769497	1.995980	2.937472	
min	4.0	11.000000	40.000000	2010.000000	1.000000	0.000000	50.000000	1.000000	1.000000	1.00000	1.000000	1.000000	1.000000	
25%	4.0	16.000000	203.000000	2011.000000	4.000000	0.000000	54.299999	9.000000	2.000000	1.00000	1.000000	1.000000	4.000000	
50%	4.0	23.000000	250.000000	2011.000000	6.000000	1.000000	57.700001	12.000000	3.000000	2.00000	5.000000	1.000000	6.000000	
75%	4.0	32.000000	528.000000	2011.000000	8.000000	1.000000	60.900002	14.000000	4.000000	4.00000	5.000000	5.000000	9.000000	
max	4.0	35.000000	756.000000	2012.000000	12.000000	1.000000	64.000000	25.000000	5.000000	4.00000	5.000000	5.000000	10.000000	

1.2

In [195…
```
share_filtered['exercise'] = np.where(share_filtered['br015'] <= 2, 1, 0)
```

In [196…
```
share_filtered.value_counts('exercise', normalize=True)
```

Out[196…
```
exercise
1    0.570384
0    0.429616
Name: proportion, dtype: float64
```

I have defined excercising as engaging in a physical activity at least once a week. One or to three times a month is not enough and could include people who just take a brisk walk to the grocery shop. Per this definition, approximately 57% of the people in the sample exercise.

1.3

In [197…
```
share_filtered['marriage'] = np.where(share_filtered['mar_stat'] <= 2, 1, 0)
```

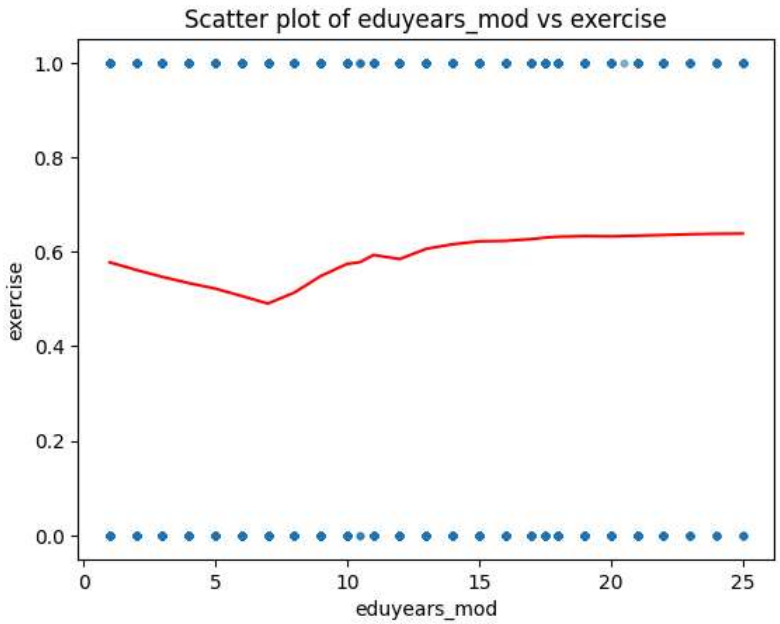
In [198…
```
share_filtered.value_counts('marriage', normalize=True)
```

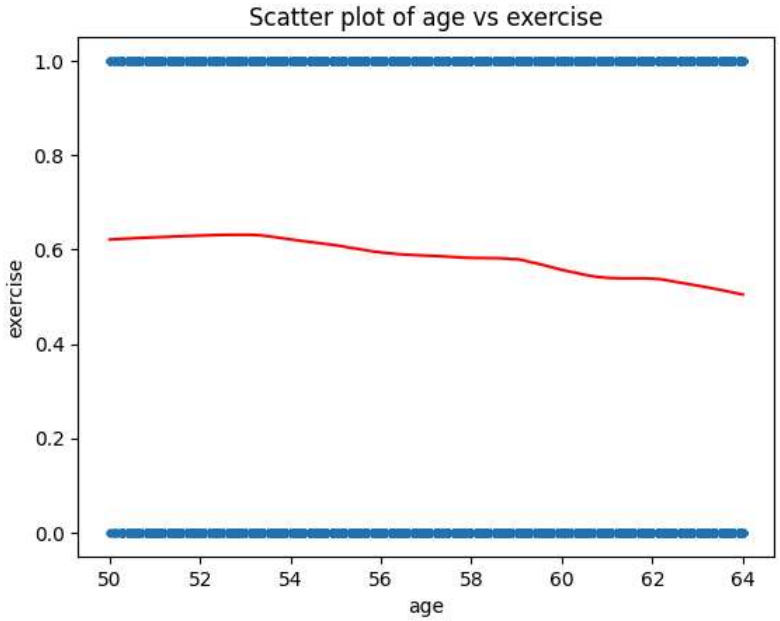
Out[198…
```
marriage
1    0.743459
0    0.256541
Name: proportion, dtype: float64
```

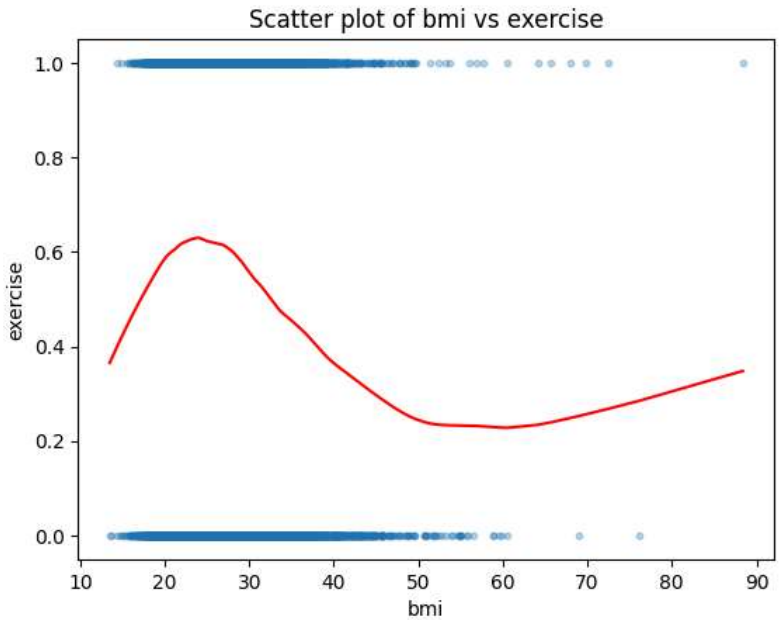
Part 2

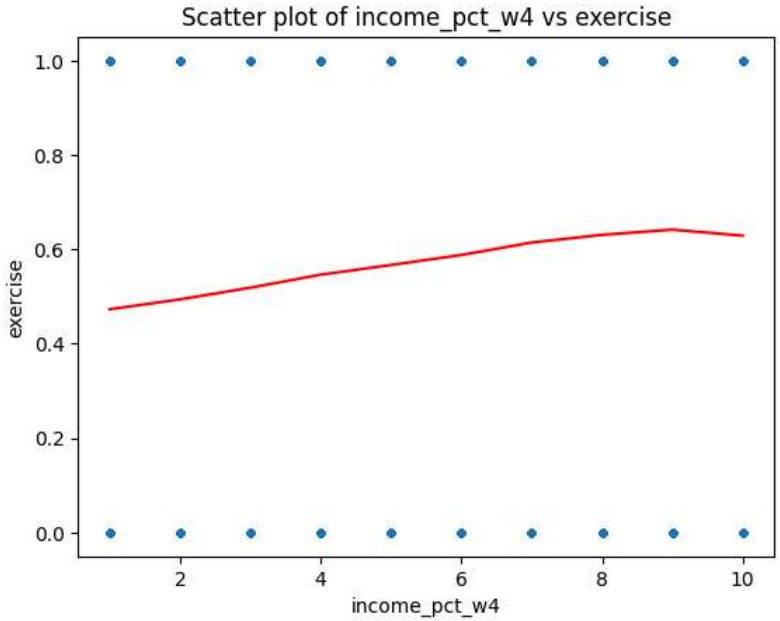
2.1

In [199…
```
for col in ['eduyears_mod', 'age', 'bmi', 'income_pct_w4']:

    lowess_fit = lowess(
        endog=share_filtered['exercise'],
        exog=share_filtered[col],
        frac=0.3   # smoothing parameter
    )

    plt.scatter(share_filtered[col], share_filtered['exercise'], alpha=0.3, s=10)
    plt.plot(lowess_fit[:, 0], lowess_fit[:, 1], color='red')
    plt.xlabel(col)
    plt.ylabel('exercise')
    plt.title(f'Scatter plot of {col} vs exercise')
    plt.show()
```


Scatter plot of eduyears_mod vs exercise
exercise
eduyears_mod
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25


Scatter plot of age vs exercise
exercise
age
1.0
0.8
0.6
0.4
0.2
0.0
50
52
54
56
58
60
62
64





2.2

In [200…

```
# Create splines
x = share_filtered["eduyears_mod"].to_numpy()
```

```
share_filtered["eduyears_mod_0_7"] = np.minimum(x, 7.0)
share_filtered["eduyears_mod_7_17"] = np.minimum(np.maximum(x - 7.0, 0.0), 17.0-7.0)
share_filtered["eduyears_mod_17"] = np.maximum(x - 17.0, 0.0)

x = share_filtered["bmi"].to_numpy()
share_filtered["bmi_0_24"] = np.minimum(x, 24.0)
share_filtered["bmi_24_56"] = np.minimum(np.maximum(x - 24.0, 0.0), 56.0-24.0)
share_filtered["bmi_56"] = np.maximum(x - 56.0, 0.0)

x = share_filtered["income_pct_w4"].to_numpy()
share_filtered["income_0_9"] = np.minimum(x, 9.0)
share_filtered["income_9"] = np.maximum(x - 9.0, 0.0)
```

I have decided to create splines for BMI, income, and years of education, as for these variables, the graphs show possible non-linearity. I have chosen to create the splines at points where the lowess curve takes a turn from being positive to negative or vice versa (ecuation, income, bmi) or in the middle of curves which have too much of a curvature (education)

2.3

In [201…

```
reg_exc_femage = pf.feols("exercise~female+age", data=share_filtered, vcov="HC1")
reg_exc_all = pf.feols("exercise~female+age+bmi_0_24+bmi_24_56+bmi_56+marriage+income_0_9+income_9+eduyears_mod_0_7+eduyears_mod_7_17+eduyears_mod_17", data=share_filtered, v
reg_exc_allcountry = pf.feols("exercise~female+age+bmi_0_24+bmi_24_56+bmi_56+marriage+income_0_9+income_9+eduyears_mod_0_7+eduyears_mod_7_17+eduyears_mod_17+C(country_mod)",

pf.etable([reg_exc_femage, reg_exc_all, reg_exc_allcountry],
          head_order="h",
          model_heads=["exercise", "exercise", "exercise"],
          labels={"Intercept": "Constant"}
)
```

Out[201…

	exercise		
	(1)	(2)	(3)
female	-0.097*** (0.006)	-0.093*** (0.006)	-0.093*** (0.006)
age	-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)
bmi_0_24		0.011*** (0.003)	0.011*** (0.003)
bmi_24_56		-0.013*** (0.001)	-0.013*** (0.001)
bmi_56		0.032*** (0.009)	0.029** (0.009)
marriage		-0.007 (0.008)	-0.001 (0.008)
income_0_9		0.016*** (0.001)	0.016*** (0.001)
income_9		-0.030** (0.010)	-0.027** (0.010)
eduyears_mod_0_7		-0.003 (0.003)	-0.001 (0.004)
eduyears_mod_7_17		0.006*** (0.001)	0.002 (0.001)
eduyears_mod_17		-0.007 (0.005)	-0.005 (0.004)
C(country_mod)[T.56]			-0.104*** (0.015)
C(country_mod)[T.203]			-0.100*** (0.015)
C(country_mod)[T.208]			-0.031 (0.018)
C(country_mod)[T.233]			0.010 (0.014)
C(country_mod)[T.250]			-0.119*** (0.015)
C(country_mod)[T.276]			0.100*** (0.023)
C(country_mod)[T.348]			-0.113*** (0.017)

Significance levels: * p < 0.05, ** p < 0.01, *** p < 0.001. Format of coefficient cell: Coefficient (Std. Error)

	exercise		
	(1)	(2)	(3)
C(country_mod)[T.380]			-0.134*** (0.017)
C(country_mod)[T.528]			0.111*** (0.017)
C(country_mod)[T.616]			-0.149*** (0.022)
C(country_mod)[T.620]			-0.196*** (0.021)
C(country_mod)[T.705]			0.016 (0.017)
C(country_mod)[T.724]			-0.217*** (0.018)
C(country_mod)[T.752]			0.063** (0.024)
C(country_mod)[T.756]			-0.020 (0.016)
Constant	1.125*** (0.047)	0.745*** (0.080)	0.850*** (0.081)
Observations	23926	23926	23926
S.E. type	hetero	hetero	hetero
R^2	0.014	0.035	0.062
Adj. R^2	0.014	0.034	0.061

Significance levels: * p < 0.05, ** p < 0.01, *** p < 0.001. Format of coefficient cell: Coefficient (Std. Error)

2.4

In model 1, the coefficient on female is -0.097 whilst for models 2 and 3, it's -0.093 (all significant at p<0.001). For model 1, this means that females (holding all else constant) are 9.7% less likely to exercise than men. For models 2 and 3, that difference in probabilities is 9.3%.

In models 1, 2, and 3, the coefficients on age are -0.009, -0.007, and -0.008 respectively (all significant at p<0.001). This means that as one gets older by a year, their predicted chance of exercising (holding all else constant) decrease by 0.9%, 0.7%, and 0.8%, respectively per the relevant model.

In models 2 and 3, the coefficients on bmi_0_24 are 0.011 (both significant at p<0.001). This means that as one increases their BMI by 1, their predicted chance of exercising (holding all else constant) increases by 1.1%, if their BMI is between 0 (not inclusive) and 24 (inclusive). The coefficients on bmi_24_56 are -0.013 (both significant at p<0.001). This means that as one increases their BMI by 1, their predicted chance of exercising (holding all else constant) decreases by 1.3%, if their BMI is between 24 (not inclusive) and 24 (inclusive). The coefficients on bmi_56 are 0.032 and 0.029 respectively (one significant at p<0.001 and the other at p<0.01). This means that as one increases their BMI by 1, their predicted chance of exercising (holding all else constant) increases by 3.2% and 2.9% respectively, if their BMI is above 56.

2.5

It appears that men tend to exercise more (or at least declare that they do so). This could be explained by differences in declaration (men might be more prone to label more activities as exercise (such as work around the house etc.)). It could also be explained by a larger portion of men in blue collar jobs.

It is interesting to see that education does not appear to have much influence on exercising besides the spline eduyears_mod_7_17 (but even there, the effects are low even with strong significance). This might be because we don't have enough observations in the other two splines. The small effect in 7_17 might be explained that as one becomes more educated, they become more aware of the health effects of a sedentary lifestyle. Thus, some people become motivated to move at least once a week.

For BMI, the biggest surprise lies in the spline bmi_56. People with a BMI above 56 are morbidly obese, so one would imagine almost no exercise compared to people with lower BMIs. However, it might be thas as people increase their BMI in the above56 spline, their health is so bad that they become forced by their caretakers to move at least a bit to minimise the negative effects on their health.

2.6

In [202…

```
reg_exc_all_nospline = pf.feols("exercise~female+age+bmi+marriage+income_pct_w4+eduyears_mod", data=share_filtered, vcov="HC1")

pf.etable([reg_exc_all_nospline, reg_exc_all],
          head_order="h",
          model_heads=["exercise", "exercise"],
          labels={"Intercept": "Constant"}
)
```

Out[202…

	exercise	
	(1)	(2)
female	-0.099***	-0.093***
	(0.006)	(0.006)
age	-0.007***	-0.007***
	(0.001)	(0.001)
bmi	-0.009***	
	(0.001)	
marriage	-0.002	-0.007
	(0.008)	(0.008)
income_pct_w4	0.014***	
	(0.001)	
eduyears_mod	0.003***	
	(0.001)	
bmi_0_24		0.011***
		(0.003)
bmi_24_56		-0.013***
		(0.001)
bmi_56		0.032***
		(0.009)
income_0_9		0.016***
		(0.001)
income_9		-0.030**
		(0.010)
eduyears_mod_0_7		-0.003
		(0.003)
eduyears_mod_7_17		0.006***
		(0.001)
eduyears_mod_17		-0.007
		(0.005)
Constant	1.152***	0.745***
	(0.051)	(0.080)
Observations	23926	23926
S.E. type	hetero	hetero
R^2	0.031	0.035
Adj. R^2	0.031	0.034

Significance levels: * p < 0.05, ** p < 0.01, *** p < 0.001. Format of coefficient cell: Coefficient (Std. Error)

Using the splines was a correct decision. The R^2 increases slightly. We also get a glimpse into the interval of education that makes the biggest difference (between 7 and 17 years) compared to the others that are not statisticall significant.

3.1

In [203…

```
reg_exc_all_lpm = pf.feols("exercise~female+age+bmi_0_24+bmi_24_56+bmi_56+marriage+income_0_9+income_9+eduyears_mod_0_7+eduyears_mod_7_17+eduyears_mod_17", data=share_filtere
reg_exc_all_logit = pf.feglm("exercise~female+age+bmi_0_24+bmi_24_56+bmi_56+marriage+income_0_9+income_9+eduyears_mod_0_7+eduyears_mod_7_17+eduyears_mod_17", data=share_filte
reg_exc_all_probit = pf.feglm("exercise~female+age+bmi_0_24+bmi_24_56+bmi_56+marriage+income_0_9+income_9+eduyears_mod_0_7+eduyears_mod_7_17+eduyears_mod_17", data=share_filt

pf.etable([reg_exc_all_lpm, reg_exc_all_logit, reg_exc_all_probit],
          head_order="h",
          model_heads=["exercise (OLS)", "exercise (Logit)", "exercise (Probit)"],
          labels={"Intercept": "Constant"}
)
```

Out[203…

	exercise (OLS)	exercise (Logit)	exercise (Probit)
	(1)	(2)	(3)
female	-0.093*** (0.006)	-0.390*** (0.027)	-0.242*** (0.017)
age	-0.007*** (0.001)	-0.031*** (0.003)	-0.019*** (0.002)
bmi_0_24	0.011*** (0.003)	0.046*** (0.011)	0.028*** (0.007)
bmi_24_56	-0.013*** (0.001)	-0.054*** (0.004)	-0.034*** (0.002)
bmi_56	0.032*** (0.009)	0.160* (0.068)	0.097* (0.040)
marriage	-0.007 (0.008)	-0.028 (0.033)	-0.017 (0.020)
income_0_9	0.016*** (0.001)	0.068*** (0.006)	0.042*** (0.004)
income_9	-0.030** (0.010)	-0.127** (0.044)	-0.079** (0.027)
eduyears_mod_0_7	-0.003 (0.003)	-0.013 (0.014)	-0.008 (0.009)
eduyears_mod_7_17	0.006*** (0.001)	0.027*** (0.005)	0.017*** (0.003)
eduyears_mod_17	-0.007 (0.005)	-0.031 (0.019)	-0.019 (0.012)
Constant	0.745*** (0.080)	1.022** (0.333)	0.635** (0.207)
Observations	23926	23926	23926
S.E. type	hetero	hetero	hetero
R^2	0.035	-	-
Adj. R^2	0.034	-	-

Significance levels: * p < 0.05, ** p < 0.01, *** p < 0.001. Format of coefficient cell: Coefficient (Std. Error)

In [204…

```
# LPM
reg_exc_all_lpm = smf.ols(
    formula=(
        "exercise ~ female + age "
        "+ bmi_0_24 + bmi_24_56 + bmi_56 "
        "+ marriage "
        "+ income_0_9 + income_9 "
        "+ eduyears_mod_0_7 + eduyears_mod_7_17 + eduyears_mod_17"
```

```python
    ),
    data=share_filtered,
).fit(cov_type="HC1")

# Logit
reg_exc_all_logit_sm = smf.logit(
    formula=(
        "exercise ~ female + age "
        "+ bmi_0_24 + bmi_24_56 + bmi_56 "
        "+ marriage "
        "+ income_0_9 + income_9 "
        "+ eduyears_mod_0_7 + eduyears_mod_7_17 + eduyears_mod_17"
    ),
    data=share_filtered,
).fit(cov_type="HC1")

# Probit
reg_exc_all_probit_sm = smf.probit(
    formula=(
        "exercise ~ female + age "
        "+ bmi_0_24 + bmi_24_56 + bmi_56 "
        "+ marriage "
        "+ income_0_9 + income_9 "
        "+ eduyears_mod_0_7 + eduyears_mod_7_17 + eduyears_mod_17"
    ),
    data=share_filtered,
).fit(cov_type="HC1")

logit_marg = reg_exc_all_logit_sm.get_margeff(at="overall")
probit_marg = reg_exc_all_probit_sm.get_margeff(at="overall")
```

```
Optimization terminated successfully.
         Current function value: 0.665647
         Iterations 5
Optimization terminated successfully.
         Current function value: 0.665641
         Iterations 5
```

In [205…

```python
logit_marg.summary()
```

Out[205…

Logit Marginal Effects

Dep. Variable: exercise

Method: dydx

At: overall

	dy/dx	std err	z	P>|z|	[0.025	0.975]
female	-0.0922	0.006	-14.464	0.000	-0.105	-0.080
age	-0.0074	0.001	-9.123	0.000	-0.009	-0.006
bmi_0_24	0.0108	0.003	4.000	0.000	0.006	0.016
bmi_24_56	-0.0128	0.001	-15.380	0.000	-0.014	-0.011
bmi_56	0.0379	0.016	2.367	0.018	0.007	0.069
marriage	-0.0067	0.008	-0.863	0.388	-0.022	0.009
income_0_9	0.0161	0.001	11.842	0.000	0.013	0.019
income_9	-0.0300	0.010	-2.888	0.004	-0.050	-0.010
eduyears_mod_0_7	-0.0030	0.003	-0.871	0.384	-0.010	0.004
eduyears_mod_7_17	0.0065	0.001	5.078	0.000	0.004	0.009
eduyears_mod_17	-0.0073	0.005	-1.590	0.112	-0.016	0.002

In [206…

```
probit_marg.summary()
```

Out[206…

Probit Marginal Effects

Dep. Variable: exercise

Method: dydx

At: overall

	dy/dx	std err	z	P>|z|	[0.025	0.975]
female	-0.0925	0.006	-14.491	0.000	-0.105	-0.080
age	-0.0074	0.001	-9.134	0.000	-0.009	-0.006
bmi_0_24	0.0108	0.003	3.992	0.000	0.005	0.016
bmi_24_56	-0.0128	0.001	-15.445	0.000	-0.014	-0.011
bmi_56	0.0372	0.014	2.635	0.008	0.010	0.065
marriage	-0.0065	0.008	-0.829	0.407	-0.022	0.009
income_0_9	0.0161	0.001	11.851	0.000	0.013	0.019
income_9	-0.0301	0.010	-2.914	0.004	-0.050	-0.010
eduyears_mod_0_7	-0.0029	0.003	-0.846	0.398	-0.010	0.004
eduyears_mod_7_17	0.0064	0.001	5.055	0.000	0.004	0.009
eduyears_mod_17	-0.0074	0.005	-1.613	0.107	-0.016	0.002

3.2

In [207…

```
share_filtered["pred_lpm"]    = reg_exc_all_lpm.predict(share_filtered)
share_filtered["pred_logit"]  = reg_exc_all_logit_sm.predict(share_filtered)
share_filtered["pred_probit"] = reg_exc_all_probit_sm.predict(share_filtered)

preds = {
    "LPM": share_filtered["pred_lpm"].astype(float),
    "Logit": share_filtered["pred_logit"].astype(float),
    "Probit": share_filtered["pred_probit"].astype(float),
}

# clip to avoid log_loss issues (and to make LPM usable as probs)
eps = 1e-15
preds_clip = {k: v.clip(eps, 1 - eps) for k, v in preds.items()}
y = share_filtered["exercise"].astype(float)

metrics = pd.DataFrame(
    {
        "R-squared": {
            "LPM": float(reg_exc_all_lpm.rsquared),
            "Logit": r2_score(y, preds["Logit"]),
            "Probit": r2_score(y, preds["Probit"]),
        },
        "Brier-score": {k: mean_squared_error(y, v) for k, v in preds.items()},
        "Pseudo R-squared": {
            "LPM": np.nan,
            "Logit": float(reg_exc_all_logit_sm.prsquared),
            "Probit": float(reg_exc_all_probit_sm.prsquared),
```

```
        },
        "Log-loss": {k: log_loss(y, v) for k, v in preds_clip.items()},
    }
).T.round(3)

metrics
```

Out[207…

	LPM	Logit	Probit
R-squared	0.035	0.035	0.035
Brier-score	0.237	0.237	0.237
Pseudo R-squared	NaN	0.026	0.026
Log-loss	0.666	0.666	0.666

It seems that there is no significant difference between the models when it comes to explanatory power.